



Data Retrieval Strategies

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Learning Objectives

- Understand TCGA's structure and scope.
- Learn how to retrieve CNV, RNA-Seq, clinical and proteomics data from TCGA.
- Compare different strategies for accessing TCGA data.
- Evaluate best-fit methods for your project context

Overview of Common File Formats

Data Type	Description	Format(s)
CNV	Copy number variations from SNP6 array or WES	SEG, gene-level TSV
RNA-Seq	Expression of transcripts per sample	FASTQ, BAM, HTSeq Counts
Clinical	Patient metadata, survival, treatment outcomes	XML, BCR TSV, JSON
Proteomics	CPTAC-linked projects, protein quantification	CSV, RPPA matrices

Access Pathways for TCGA Data

- **GDC Data Portal:** Official and comprehensive source.
- **UCSC Xena Browser:** Intuitive interface + easy clinical integration.
- **TCGAbiolinks** (R/Bioconductor): Programmatic, automated access.
- **GDC API / gdc-client:** Bulk download, reproducible pipelines

Strategy 1 – Access via **GDC Data Portal**

➤ **Pros:**

- Full metadata and raw/processed files
- Controlled access tier allows sensitive clinical data
- Granular filters (project, data type, tissue, program etc.)

Strategy 1 – Access via **GDC Data Portal**

➤ **Cons:**

- Steep learning curve for beginners
- Requires dbGaP approval for controlled tier
- File size and metadata can be overwhelming

Strategy 2 – Access via UCSC Xena Browser

➤ Pros:

- Easy filtering of clinical and expression data
- Ready-to-analyze matrices: CNV, RNA-Seq, survival
- Plot survival and genomic events visually

Strategy 2 – Access via UCSC Xena Browser

➤ Cons:

- Only preprocessed data
- Some granularity lost (no BAM/FASTQ)
- Limited flexibility for deep re-analysis

Strategy 3 – Access via **TCGAbiolinks (R)**

➤ **Pros:**

- Integrated with R ecosystem
- Automates querying, downloading, and formatting
- Supports statistical and differential expression pipelines

Strategy 2 – Access via **TCGAbiolinks (R)**

➤ **Cons:**

- Requires R programming experience
- Some preprocessing steps must be customized manually

Extracting CNV Data from TCGA

- Use **GDC portal** to filter for CNV (SNP array/WES-derived)
- **Files:** SEG format (segment-level) or gene-level matrices
- Xena offers downloadable gene-level CNV calls

Extracting **RNA-Seq** Data from TCGA

- **Levels:** Raw FASTQ > BAM > HTSeq counts > normalized matrices
- **GDC** = All levels (controlled access for FASTQ)
 - **Xena** = Pre-normalized TPM/FPKM/Counts
 - **TCGAbiolinks** = Flexible querying and transformation

Extracting Clinical Data from TCGA

- **GDC:** Clinical (demographics, vital status, therapies) via JSON or TSV.
- **Xena:** Clean survival tables with quick filters.
- **TCGAbiolinks:** Structured clinical fields with easy parsing.

Extracting Proteomics Data from TCGA-CPTAC

- Available for select cancer types via CPTAC portal
- Reverse Phase Protein Array (RPPA) = semiquantitative matrix
- Available via **Xena** and **LinkedOmics** (for integration)

Which Access Route Should You Choose?

Tool	Ideal For	Format
GDC Portal	Raw data, detailed filters	FASTQ, BAM, TSV
Xena Browser	Quick analysis + clinical integration	Matrix, VIZ
TCGAbiolinks	Reproducible pipelines in R	R Objects
GDC API	Bulk download, scripted automation	JSON, files

Summary of TCGA Data Strategy

- TCGA provides rich, **multi-modal** cancer data.
- Choose your access method based on your **skills** and **goals**.
- Understand **file types** and **preprocessing levels**.
- Always **document** data provenance for reproducibility.

Practice Discussion

Q: Your team is analyzing **RNA-Seq** and **CNV** data for breast cancer survival analysis.

Which **TCGA access strategy** will you choose and **why?**

Conclusion

- **Biological data access** is essential for modern cancer research. TCGA equips you with extensive and multi-layered datasets, and knowing how to access and handle them is what transforms curiosity into discovery.

Further Reading/ References

- **TCGA GDC Portal Tutorials:** <https://gdc.cancer.gov/access-data/gdc-data-portal-user-guide>
- **TCGAbiolinks Workshop Slides:**
- <https://bioconductor.org/packages/release/bioc/vignettes/TCGAbiolinks/>
- **UCSC Xena Tutorials:** <https://ucsc-xena.gitbook.io/project/>
- **NCI Genomic Data Commons Docs:** <https://docs.gdc.cancer.gov/>
- **LinkedOmics Portal:** <http://www.linkedomics.org/login.php>

Coming up next...

Retrieving data from TCGA via the GDC portal:

- ✓ Analysis center
- ✓ Cohort builder
- ✓ Repository
- ✓ GDC data transfer tool